

Follow one message from your phone to your friend's.

Every chapter of the 4-hour course, redrawn as diagrams, step-throughs and flashcards. Colours stay the same everywhere: each one is an OSI layer.

11 chapters 7 layers 66 interview flashcards Source: [the video](#) + Kunal's notes

The packet journey

● Layer 1 · Physical

YOU

L7 ● Application

L6 ● Presentation

L5 ● Session

L4 ● Transport

L3 ● Network

L2 ● Data Link

L1 ● **Physical**

INTERNET

ISPs · undersea cables

R1

R2

R3

hop by hop, TTL -1 each hop

FRIEND

L7 ● Application

L6 ● Presentation

L5 ● Session

L4 ● Transport

L3 ● Network

L2 ● Data Link

L1 ● Physical

0110 1001 0011 1010 0101 ...

07/15 Physical sends bits

The frame becomes 0s and 1s, sent as electrical, light or radio signals.



How to read the tags. Plain text is from the video. **Extra context** boxes add facts the video skipped. **Instructor slip** boxes (orange) correct something the video got wrong, so you learn the right version.

How the internet started

A network is computers connected together. The internet is a network of those networks, spread across houses, cities and countries.

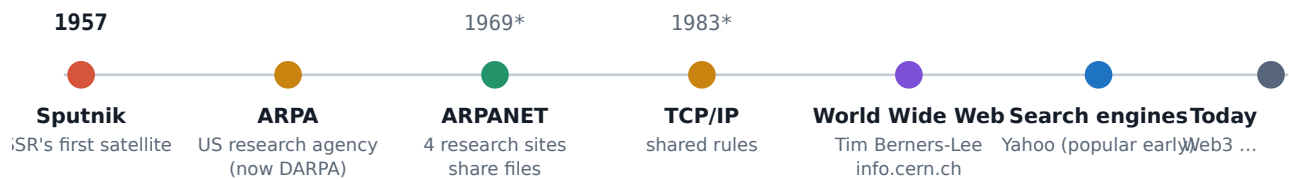
COMPUTER NETWORK

Two or more devices connected so they can share data.

INTERNET

A global collection of connected computer networks. It runs mostly through cables, including cables under the sea.

Timeline



* dates marked with a star are extra context; the video gives only 1957.

! Instructor slip. The video names MIT, Stanford, UCLA and Utah as ARPANET's first four sites, running TCP. The first four were really **UCLA, SRI (Stanford Research Institute), UC Santa Barbara and Utah**. ARPANET started on a protocol called NCP and switched to TCP/IP in 1983. The "COMPUTER = Commonly Oriented Machine..." full form is a school joke, not a definition.

Why protocols exist

Different data needs different rules, and everyone has to follow the same rules or apps can't understand each other. Those rules are

protocols

EMAIL

Follow set steps

Send to a server, the receiver downloads it, then reads it.

VIDEO CALL

Speed first

Dropping a few frames is fine.

SECRET DOCUMENTS

100% must arrive

No data may be lost.

WORLD WIDE WEB

Linked documents

Researchers wanted papers that link to other papers. WWW gives every document a URL and links them with hyperlinks; the pages live on web servers. The first website is still up at info.cern.ch.

WHO MAKES THE RULES

Internet Society + RFCs

New ideas are proposed as an **RFC (Request for Comments)**. Anyone can submit one. Extra context: the IETF, under the Internet Society, writes the standards.

Interview questions · 5

BASIC

Internet vs World Wide Web?

The internet is the network of networks. The WWW is a service on it: linked pages with URLs, served by web servers.

WHY

Why was ARPANET built?

ARPA's research sites were far apart and needed to share files and talk. It came out of the US response to Sputnik (1957).

WHY

Why do we need protocols?

Different data needs different delivery rules, and every system must use the same rules to understand each other.

BASIC

What is an RFC?

A Request for Comments: a document proposing a standard or feature for the internet.

SCENARIO

Two chat apps use different message formats. Can they talk?

No. Without a shared protocol neither can read the other's data.

Exam revision

—

ARPA Advanced Research Projects Agency

WWW Tim Berners-Lee

RFC Request for Comments

Sputnik **1957**

internet ≠ WWW

Clients, servers, IP addresses and ports

An IP address finds the device. A port number finds the app on that device. This chapter sets up the ideas the rest of the course goes deeper into.



On localhost, one computer is both client and server.

Demo in the video: Inspect → Network → refresh google.com shows dozens of GET/POST requests, status 200 and many file types.

TCP

Must arrive intact

Transmission Control Protocol. Makes sure data reaches its destination uncorrupted.

UDP

Fast, may lose some

User Datagram Protocol. Used for video calls.

HTTP

Web pages

HyperText Transfer Protocol. Rules for how browsers request and servers reply.

Data never travels in one piece. It is cut into small chunks called

packets

IP address: the phone book

IP ADDRESS

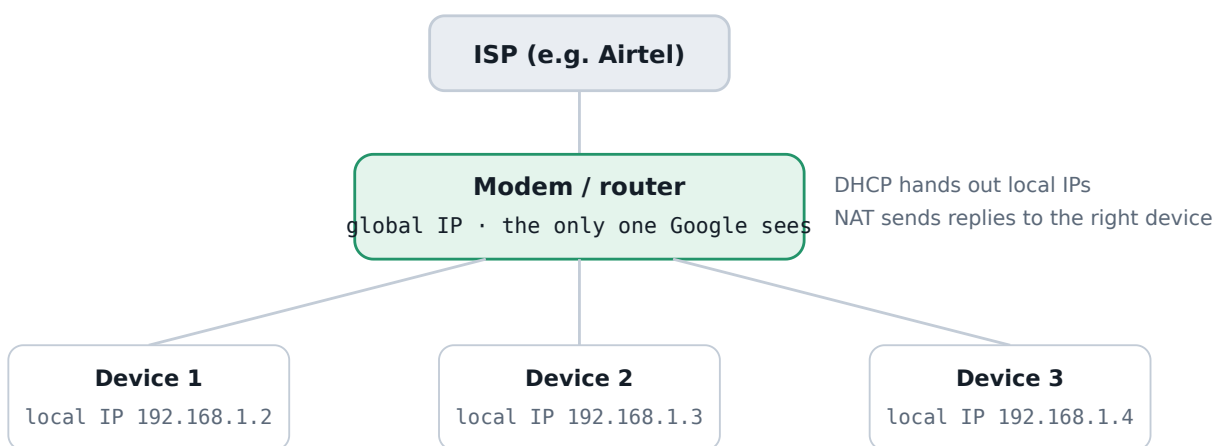
A number that identifies every device on the internet. Format $x.x.x.x$, each x from 0 to 255.

ANALOGY FROM THE VIDEO

You tap "John" and your phone dials the number saved under that name. Typing google.com works the same way: the name is turned into an IP address.

`curl ifconfig.me -s` shows your public IP.

One router, many devices



Local IPs here are example values. The video's point: every device in the house shares one global IP.



Instructor slip. NAT is said once as "network access translator". The real name is **Network Address Translation** (Chapter 10).

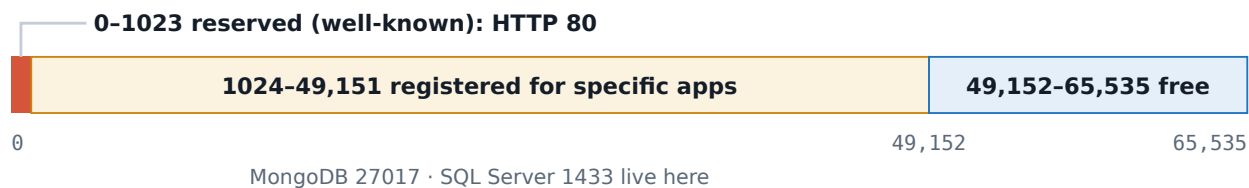
Port numbers 34:23

PROBLEM

The reply reached your laptop. But is it for Chrome, MongoDB, your own server or a game chat? All of them share the laptop's IP.

ANSWER

IP address → device. Port → application. Ports are 16-bit numbers, so there are $2^{16} = 65,536$ of them (0-65,535).



Drawn to scale. The video says registered ports run "1024 till something like 49,152"; the official split is 49,151 / 49,152.

Speed and links

UNIT	BITS PER SECOND
1 kbps	$10^3 = 1,000$
1 Mbps	$10^6 = 1,000,000$
1 Gbps	10^9

REMEMBER

Mbps is megabits, not megabytes.
Divide by 8 for MB/s. Upload = you send;
download = you receive.
Guided links follow a wire (Ethernet,
fibre). **Unguided** links don't (Wi-Fi,
Bluetooth).

Interview questions · 6

BASIC

IP address vs port?

IP identifies the device; the port identifies the application on it.

WHY

Why do all home devices look like one device to Google?

They share the router's single global IP. DHCP gives them local IPs and NAT routes replies back.

BASIC

How many ports exist?

65,536 (0–65,535), because a port is 16 bits.

SCENARIO

Can your app take port 80?

Not normally. 0–1023 are reserved; 80 belongs to HTTP.

COMPARE

Is 100 Mbps the same as 100 MB/s?

No. It's megabits; that's about 12.5 MB/s.

SCENARIO

You game and browse at once. How does a reply reach the game?

By its destination port number.

Exam revision

octet **0-255**

ports **16-bit**

reserved **0-1023**

HTTP **80**

MongoDB **27017**

SQL Server **1433**

global vs local IP

guided vs unguided

The physical internet

"The internet is not in the cloud, it's under the sea." Countries are joined by fibre cables on the ocean floor, because cables are faster than satellites.

SUBMARINECABLEMAP.COM

India's links

Chennai and Kochi connect south (Sri Lanka); Mumbai connects to Dubai, Oman, UAE and on to Singapore and Malaysia.

ONE CABLE

≈ 28,000 km

Japan, South Korea, China, Malaysia, India, UAE, Israel, Italy, UK.

SHARKS?

Not a threat

Cables are heavily protected and buried. Google owns many of them.

WIRED

Optical fibre

Coaxial cable

Ethernet cable

WIRELESS

Bluetooth (short range)

Wi-Fi

3G · 4G/LTE · 5G (long range)

LAN, MAN, WAN

WAN · across countries · optical fibre (SONET, Frame Relay)

MAN · a city

LAN · home

Wi-Fi, Ethernet switch, adapters

LAN · office

can be 10,000 computers

MAN · another city

LAN · college

LAN · shop

LAN means one area, not a few computers. SONET = Synchronous Optical Networking (data over fibre, long distances). Frame Relay connects a LAN to the wider network.

MODEM

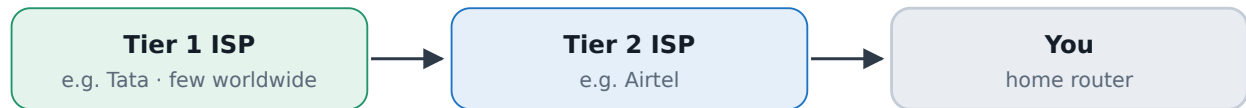
Digital ↔ analog

Turns your computer's digital data into a signal for telephone lines, and back again at the other end. (MODulator-DEModulator.)

ROUTER • LAYER 3

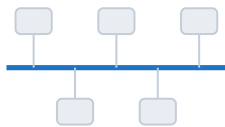
Routes packets by IP

Connects LANs and WANs and decides where each packet goes.



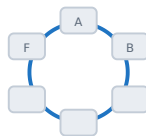
Each smaller ISP buys access from a bigger one.

Topologies 55:47



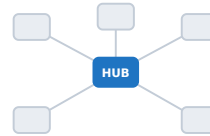
Bus

One backbone. If it breaks, everything fails; only one computer sends at a time.



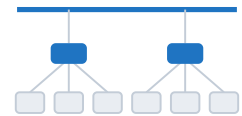
Ring

A to F may pass through B and C. One broken cable stops it; many unneeded hops.



Star

Everything goes through one central device. If it fails, the network goes down.



Tree

Stars hanging off a bus. A bit more fault tolerant.



Mesh

Every computer wired to every other. Expensive and hard to scale. Extra context: n devices need $n(n-1)/2$ links.

Interview questions · 6

WHY

Why undersea cables and not satellites?

Cables are faster.

COMPARE

LAN vs MAN vs WAN?

Area covered: a building, a city, countries. LAN size is about area, not device count.

COMPARE

Modem vs router?

A modem converts digital and analog signals; a router forwards packets between networks by IP (Layer 3).

BASIC

What is a Tier 1 ISP?

A top-level provider other ISPs connect through, e.g. Tata in India.

CONCEPT

Which topology has a single point of failure?

Star (the central device). Bus also dies if the backbone breaks.

SCENARIO

One broken cable must never cut anyone off. Which topology?

Mesh, at a higher cost.

Exam revision

SONET = Synchronous Optical Networking

Frame Relay: LAN → WAN

modem = digital↔analog

router = **L3**

5 topologies + limits

The OSI model, layer by layer

OSI (Open Systems Interconnection) splits communication into 7 layers. Each layer does one job and trusts the layer below to do its own. The video calls it the most important interview topic.

The Amazon analogy



You only ever see the app (L7). Everything else happens underneath, like WhatsApp hiding the internet from you.

Layer explorer

Pick a layer to see what it does, its data unit, addressing, protocols and devices.

7 Application Data

6 Presentation Data

5 Session Data

4 Transport Segment

3 Network Packet

2 Data Link Frame

● Layer 7 Application

The layer users interact with, through apps.

- You send messages, files and emails through an app
- Lives in software, on your device
- Hands the data down to presentation

DATA UNIT
Data / message

ADDRESS
— (names, via DNS)

PROTOCOLS
HTTP, FTP, Telnet,
DNS, SMTP

DEVICES
Software on end
devices

1 Physical

Bits

ANALOGY

The Amazon app
where you order

i Extra context. The video lists layers top-down ("first is application"). Exams number them bottom-up: Physical = 1, Application = 7. Mnemonic, top-down: **All People Seem To Need Data Processing**.

Encapsulation: what gets added on the way down

Application

DATA

Transport segment

ports-seq-chk

DATA

Network packet

src/dst IP

TCP/UDP hdr

DATA

Data link frame

MACs

IP hdr

TCP/UDP hdr

DATA

trailer

Physical bits

0110 1001 0011 1010 0101 1100 ...

Going down

each layer wraps the one above in its header.

Going up

each layer removes its own

Layer order and what each layer adds are from the video; the exact header names are standard extra context.

! Instructor slip. At 1:22:33 the video says physical (MAC) addressing decides "which application" gets the data. The **port number** (Layer 4) picks the application. The MAC address picks the **device's network interface** on the local network.

All 7 layers at a glance

#	LAYER	JOB	DATA UNIT	ADDRESS	PROTOCOLS	DEVICES
● 7	Application	Apps users touch	Data	—	HTTP, FTP, Telnet, DNS, SMTP	software
● 6	Presentation	Translate, encrypt, compress	Data	—	SSL	—
● 5	Session	Open, keep, close sessions; login checks	Data	—	extra: NetBIOS, RPC	—
● 4	Transport	Segments, flow & error control, right app	Segment	Port	TCP, UDP	end devices
● 3	Network	IP addressing, routing, best path	Packet	IP (logical)	IP	Router
● 2	Data link	MAC addressing, frames, media access, error detection	Frame	MAC (physical)	ARP, Ethernet	Switch, bridge
● 1	Physical	Bits to electrical, light or radio signals	Bits	—	cables, radio	Hub, repeater

Interview questions · 8

BASIC

Name the OSI layers.

Application, Presentation, Session, Transport, Network, Data Link, Physical.

CONCEPT

What does the presentation layer do?

Translation (e.g. ASCII → EBCDIC), encryption with SSL, and lossy or lossless compression.

CONCEPT

What does the session layer do?

Sets up, maintains and ends sessions, including authentication and authorization (e.g. a Flipkart login until you log out).

CONCEPT

Three jobs of the transport layer?

Segmentation, flow control, error control.

COMPARE

Logical vs physical addressing?

Logical = IP at Layer 3, end to end. Physical = MAC at Layer 2, per local link.

BASIC

Which layer are router, switch, hub?

Router L3, switch/bridge L2, hub/repeater L1.

SCENARIO

A page shows garbled symbols. Which layer's job?

Presentation (translation/encoding).

SCENARIO

Server sends 40 Mbps, client takes 20 Mbps. What prevents loss?

Flow control at the transport layer.

Exam revision

Data → Segment → Packet → Frame → Bits

Port · IP · MAC

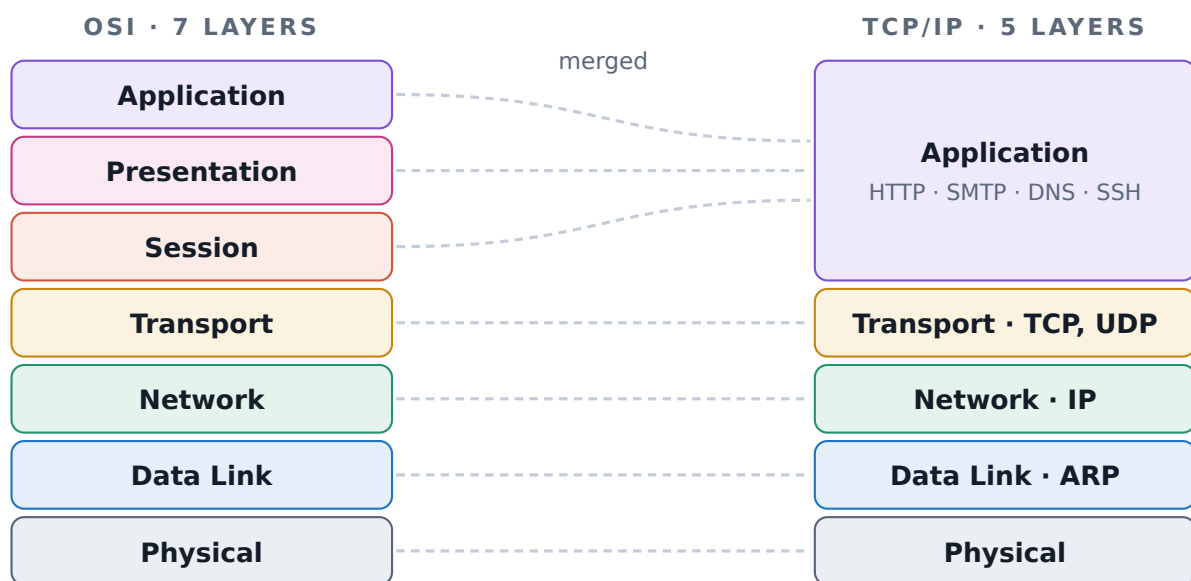
SSL at L6

ASCII → EBCDIC

router L3 · switch L2 · hub L1

The TCP/IP model

The model the real internet runs on, also called the Internet Protocol Suite. It keeps OSI's lower layers and folds the top three into one Application layer. OSI is for theory; TCP/IP is used in practice.



Extra context: some books show TCP/IP with 4 layers, joining data link and physical. This course uses 5.

POINT	OSI	TCP/IP
Layers	7	5 (4 in some books)
Top layers	Application, Presentation, Session kept apart	Merged into one Application layer
Used for	Theory and concepts	The working internet
Origin	A standard for how systems talk	US ARPA work
Named after	Protocol-independent	Its two key protocols, TCP and IP

i Extra context. DHCP is formally an application-layer protocol running on UDP. The video talks about it under the data link layer because it gives addresses to devices joining a local network.

Interview questions · 4

COMPARE

OSI vs TCP/IP?

TCP/IP merges OSI's application, presentation and session into one layer. TCP/IP is practical; OSI is conceptual.

WHY

Why the name TCP/IP?

It's built around TCP (transport) and IP (network).

CONCEPT

Where does encryption happen in TCP/IP?

Inside the application layer (e.g. HTTPS), since presentation is merged into it.

CONCEPT

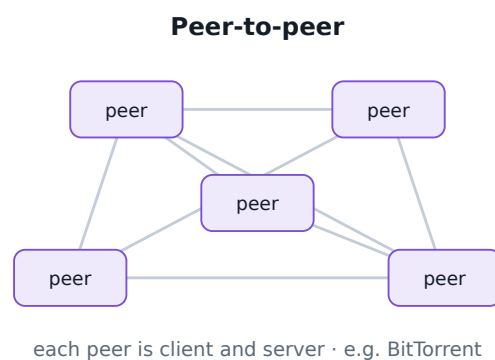
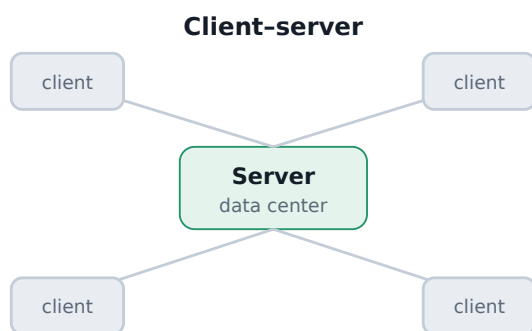
4 or 5 layers?

Both appear; 5 keeps data link and physical separate.

Application layer: architectures, devices, protocols, HTTP

Where users meet the network. Apps connect either through a central server or directly as peers, and every exchange follows a protocol.

Client-server vs peer-to-peer



SERVERS NEED

- A reliable IP that clients can reach
- High availability
- Big companies use **data centers**: static IPs, always-on links, high upload speed
- Overload crashes sites: exam results day, the PS5 pre-order launch

P2P

- No central server; devices connect directly
- Scales very fast, decentralised
- Example: BitTorrent seeding
- Hybrids exist: P2P plus a central database

ping 1:36:26

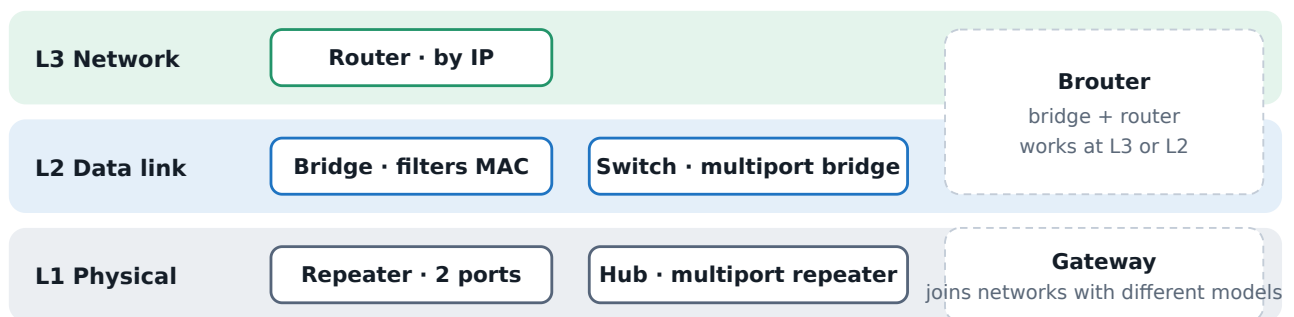
```
$
ping google.com
64 bytes from 142.250.x.x: icmp_seq=0 ttl=60
time=21 ms
64 bytes from 142.250.x.x: icmp_seq=1 ttl=60
time=21 ms
...
11 packets transmitted, 11 received,
0% loss
```

READ IT LIKE THIS

- **64 bytes:** each packet's size
- **icmp_seq:** sequence number keeps order
- **ttl:** hops left before the packet is dropped (Ch 10)
- **time:** round-trip time, "ping time"

Output reconstructed for illustration; IPs differ because Google has many servers.

Networking devices 1:39:52



REPEATER

Regenerates a weakening signal bit by bit to its original strength. It does **not** amplify.

HUB

Sends everything to every port: no filtering, one collision domain for all hosts. **Active** hubs have their own power and clean, boost and relay; **passive** hubs just pass it on.

HUB	SWITCH
Layer 1	Layer 2
Multiport repeater	Multiport bridge
Floods all ports	Forwards by MAC
One collision domain	One per port (extra)
No error check	Error-checks first

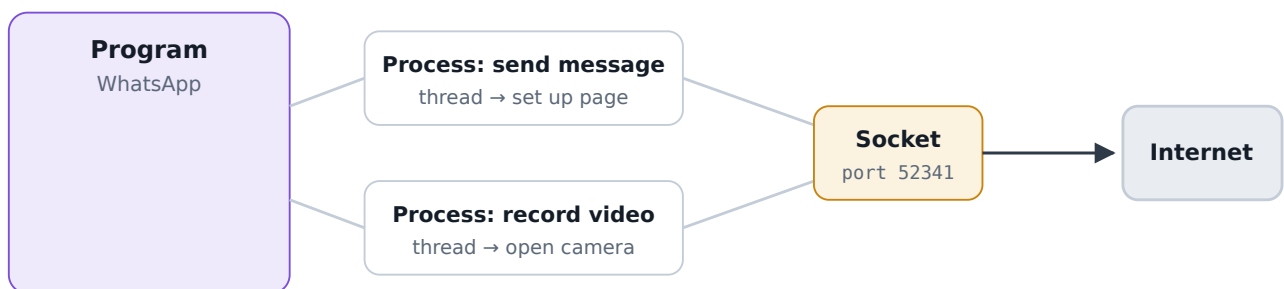
SWITCH	ROUTER
Layer 2	Layer 3
MAC addresses	IP addresses
Inside a LAN	Between networks
Frames	Packets

Application protocols 1:43:05

PROTOCOL	WHAT IT DOES	PORT	RUNS ON
HTTP	Web pages between browser and server	80	TCP
HTTPS	HTTP, secured	443 extra	TCP
DHCP	Gives devices IP addresses	67/68 extra	UDP
FTP	File transfer (rare now; HTTP does it)	20/21 extra	TCP
SMTP	Sends email	25/587 extra	TCP
POP3 / IMAP	Receive email	110 / 143	TCP
SSH	Log into another computer's terminal (cloud servers)	22 extra	TCP
VNC	Graphical remote control	5900 extra	TCP
Telnet	Remote terminal, no encryption; <code>telnet hostname</code>	23	TCP

! **Name check.** SSH is **Secure Shell** (some notes say "Secure Socket Shell"). DHCP is **Dynamic Host Configuration Protocol** (said once as "control" in the video).

Program, process, thread, socket



Program = the app. Process = a running instance or feature. Thread = a lighter unit doing one job. Socket = the software interface between a process and the internet (like the wall socket your landline plugs into). The port number shown is an example.

EPHEMERAL PORTS

Five Chrome tabs, one IP. Each instance gives itself a **random temporary port**, freed when it's done. These live on the client.

SERVER PORTS

Servers need **fixed, well-known ports** (80 for HTTP) because clients must know where to knock.

HTTP 1:53:12

WHAT

A client-server protocol: how a client asks for data and how the server replies.

RUNS ON

TCP, so every byte of the page arrives.

MEMORY

Stateless: the server forgets you between requests. Cookies fix that.

<https://www.google.com/images/branding/logo.png?type=i>

● protocol ● domain ● path to the file ● query after ? (YouTube uses ?v=videoid)

METHOD	TELLS THE SERVER
GET	Give me data (a page, a video)
POST	Take this data (a sign-up form)
PUT	Put data at this location
DELETE	Remove this data

HEADER	MEANING
Accept	Response types I'll take
Accept-Language	Preferred language
Connection	keep-alive or close
Keep-Alive	timeout, max requests
Set-Cookie	Response: store this cookie

Status codes 2:04:44

1XX

Informational
connection info

2XX

Success
200 OK

3XX

Redirect
moved
elsewhere

4XX

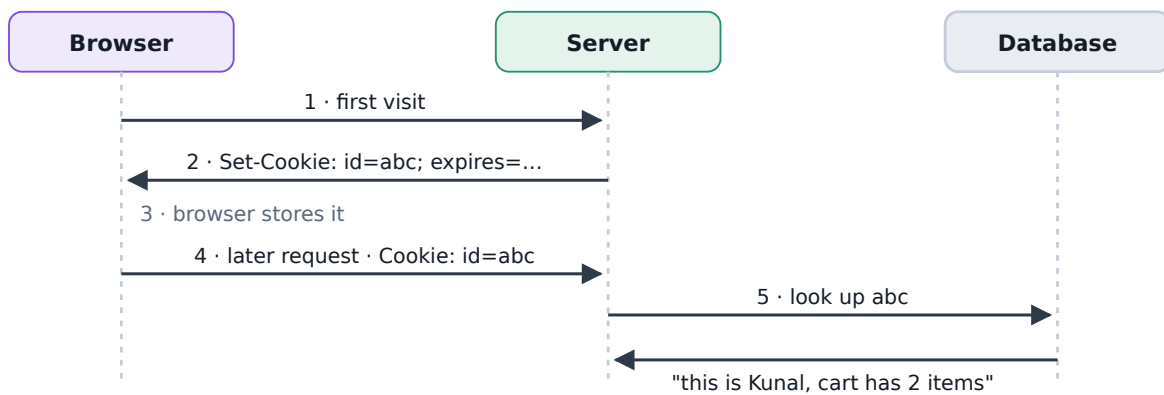
Client error
400 bad request ·
404 not found

5XX

Server error

500 internal error

Cookies 2:06:30



This is how Amazon keeps you logged in even though HTTP is stateless. **Third-party cookies** are set by sites you didn't visit, e.g. an ad embedded on flipkart.com, and can track you across sites.

COMPARE

Client-server vs P2P?

A central server answers clients, vs every node acting as both client and server.

BASIC

What does ping measure?

Round-trip time to a host; it also shows packet size, sequence, TTL and loss.

COMPARE

Hub vs switch?

Hub (L1) floods every port, one collision domain. Switch (L2) forwards by MAC and error-checks.

CONCEPT

Does a repeater amplify?

No, it regenerates the signal bit by bit.

COMPARE

Process vs thread?

A process is a running instance of a program; a thread is a lighter unit inside it doing one job.

BASIC

What is a socket?

The software interface between a process and the internet, tied to a port.

CONCEPT

What are ephemeral ports?

Temporary random client-side ports for app instances, freed after use.

WHY

Why does HTTP use TCP?

Pages must arrive complete and in order.

CONCEPT

HTTP is stateless. How do sites remember you?

Cookies sent back with every request.

SCENARIO

You renamed a page and users get 404. Whose error class?

Client error (4xx): the requested URL doesn't exist any more.

Exam revision

—

HTTP 80 · Telnet 23 · POP3 110

GET POST PUT DELETE

1xx-5xx

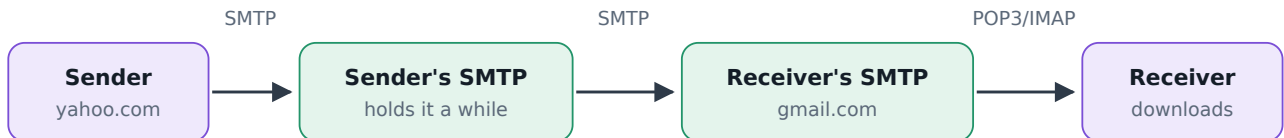
200 · 400 · 404 · 500

HTTP stateless, over TCP

third-party cookies

How email works

SMTP sends. POP3 and IMAP receive. All three run over TCP because an email must arrive whole.



Same provider on both sides? The middle hop is skipped. Receiver offline? The sending server retries for a few days.

POP3 · PORT 110

1 Authorize

:
username,
password

2 Transact

:
download
new
mail

3 Update

:
server
applies
deletes,
session
closes

Folders like Sent and Drafts don't sync.
Delete from the server and the mail lives
on one device only.

IMAP

Mail stays on the server; every device
keeps a synced copy, folders included.
Delete on your iPhone and it's gone from
your Mac and iPad too.

`nslookup -type=mx gmail.com` lists a
domain's mail (MX) servers.

POINT	POP3	IMAP
Mail lives	On the device (server copy optional)	On the server
Many devices	Poor	Built for it
Folder sync	No	Yes
Port	110	143 / 993 (extra)

Interview questions · 4

BASIC

Which protocol sends and which receive?

SMTP sends; POP3 and IMAP receive.

WHY

Why TCP for email?

Email must arrive complete.

SCENARIO

Receiving mail server is down. What happens?

The sending server keeps retrying for days, then reports non-delivery.

SCENARIO

Deletes must sync across phone and laptop. Which protocol?

IMAP.

DNS: the internet's phone book

DNS (Domain Name System) turns names like google.com into IP addresses. It's split across many servers so no single database can take the whole internet down.

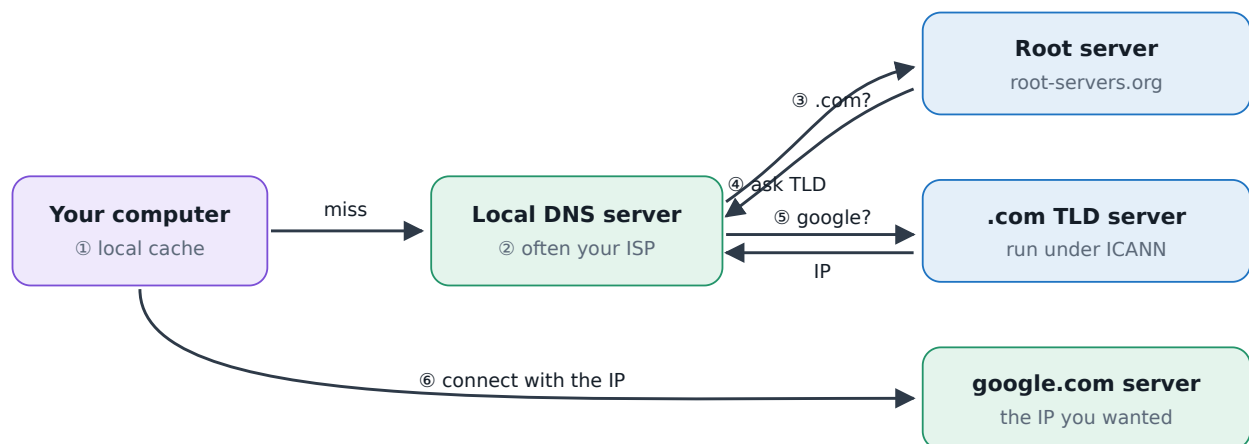
mail.google.com

• subdomain

• second-level domain

• top-level domain (TLD)

Resolving google.com



1 Local cache.

Sites you've visited are remembered on your machine.

2 Local DNS server

,
often
your
ISP's.
Your
ISP
sees
every
site
you
look
up,
even
in
incognito.

3 Root server

,
the
first
stop
after
local.
It
points
to
the
right
TLD.

4 TLD server

(.com,
.org,
.edu,
or
.in,
.uk,
.us)
returns

the

IP

5 Your computer

connects to google.com's server.



Extra context. In full detail the TLD server points to google.com's **authoritative name server**, which returns the final IP. DNS runs on **UDP port 53**.

DOMAINS

You can't buy one, only rent it: you pay GoDaddy, GoDaddy pays ICANN.

OWN TLDS

Google has `.google`; the video says Amazon has `.fire`.

TRY IT

`dig google.com`: domain information groper, a DNS lookup tool.

Interview questions · 4

WHY

Why isn't DNS one big database?

It would be a single point of failure, overloaded, with nowhere sensible to put it.

BASIC

Break down mail.google.com.

mail = subdomain, google = second-level domain, com = TLD.

BASIC

TCP or UDP for DNS?

UDP, because it's fast.

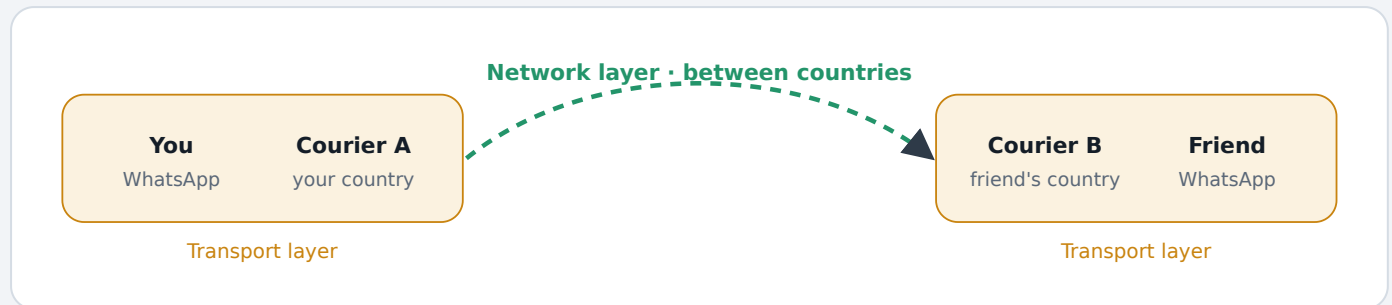
SCENARIO

You moved your site to a new IP but some users hit the old one. Why?

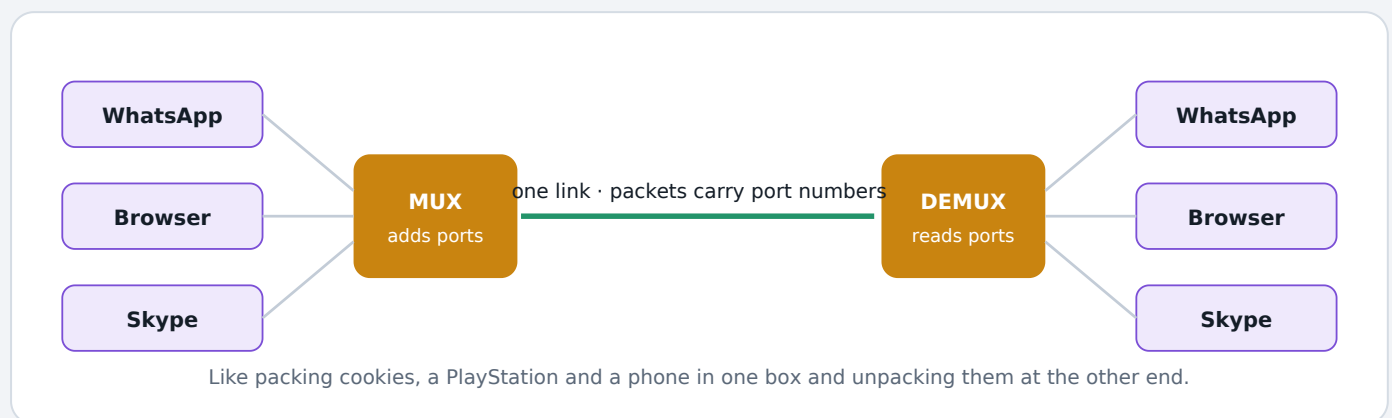
Their local DNS cache still holds the old IP.

Transport layer: getting data to the right app

The network layer moves data **between devices**. The transport layer moves it **inside a device**, between the network and the right application, and makes delivery reliable when needed.



Multiplexing and demultiplexing



Three tools for reliable delivery

CHECKSUM · DETECTS CORRUPTION

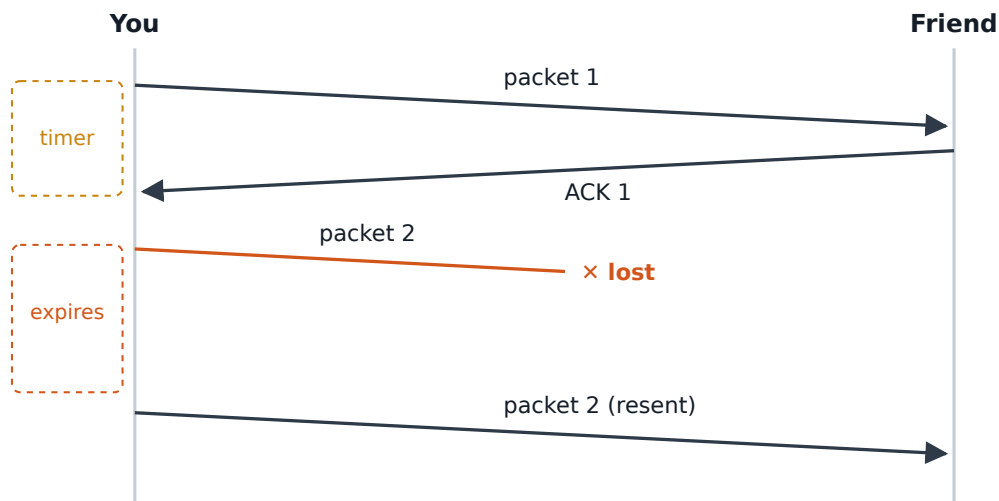
Sender computes a value from the data and attaches it. Receiver recomputes it the same way. Different value → corrupted.

TIMER · DETECTS LOSS

Start a timer on send; stop it when the ack arrives. Timer expires → resend. This is the **retransmission timer**.

SEQUENCE NUMBER · FIXES ORDER & DUPLICATES

Every segment is numbered, so a resent copy of #2 is spotted and dropped, and "my Hi Kunal name is" is put back in order.



! Instructor slip. The video calls a checksum "a random string". It isn't random: it's calculated from the data, which is why the receiver can recompute and compare it.

CONGESTION CONTROL

Too many packets for the link, like too many cars at one gate, means some get lost. The transport layer slows the send rate using congestion-control algorithms **built into TCP**.

UDP • User Datagram Protocol 2:54:00

PERSONALITY

- Connectionless: just sends
- May not arrive, may change, may be out of order
- Has a checksum, but ignores errors it finds
- No feedback, so it's **fast**
- Used for video calls, gaming, DNS

UDP header • 8 bytes

Source port 2 bytes	Destination port 2 bytes
Length 2 bytes	Checksum 2 bytes
DATA • up to $2^{16} - 8$ bytes	

Watch packets live: `sudo tcpdump -c 5`. Extra: real maximum data is 65,507 bytes over IPv4.

TCP • Transmission Control Protocol 3:02:05

CONNECTION-ORIENTED

Handshake first, then data.

RELIABLE

Error control, retransmits lost data, keeps order.

CONGESTION CONTROL

Slows down when the network is busy.

FULL DUPLEX

Both sides can send at the same time.

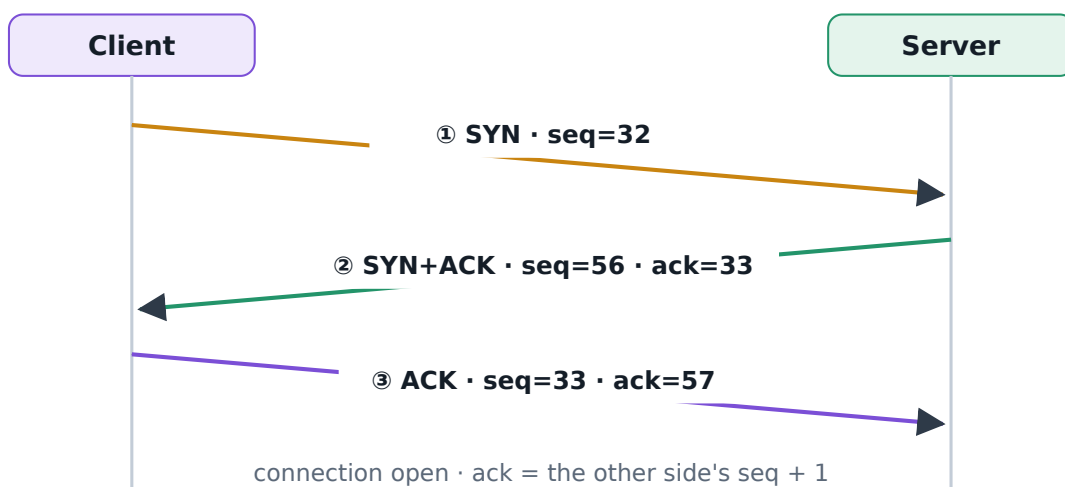
TWO ENDPOINTS

One connection joins exactly two computers.

USED BY

HTTP, SMTP, POP3, IMAP, file transfer.

The 3-way handshake 3:09:10



IN PLAIN WORDS

- 1 "Can I connect?"
- 2 "Sure, me too."
- 3 "OK, connected."

WHY RANDOM SEQUENCE NUMBERS?

Predictable numbers would let attackers guess them and hijack connections. Random numbers are for **security**.

Other flags: **RST** reset, **FIN** finish.

! Instructor slips. The server's number (56) isn't "math on the client's 32"; each side picks its **own random** starting number, and only the ack is $32 + 1$. The third message carries **only ACK**, not SYN. The video says "85" once before settling on 56.

TCP vs UDP

POINT	TCP	UDP
Connection	3-way handshake first	None
Delivery	Guaranteed	Maybe
Order	Kept (sequence numbers)	Not kept
Errors	Detects and recovers	Detects, ignores
Congestion control	Yes	No
State	Stateful	Stateless
Speed	Slower	Faster
Header	20+ bytes (extra)	8 bytes
Used by	Web, email, files	Video calls, games, DNS

COMPARE

Transport vs network layer?

Network moves data between devices; transport moves it between the network and the right app on a device.

CONCEPT

What is demultiplexing?

Splitting incoming data to the right apps using destination port numbers.

CONCEPT

How are corruption, loss and duplicates handled?

Checksum, retransmission timer, sequence numbers.

PROCESS

Explain the 3-way handshake.

SYN (seq x) → SYN-ACK (seq y, ack x+1) → ACK (seq x+1, ack y+1).

WHY

Why is UDP faster?

No handshake, no acks, no retransmission, no ordering.

BASIC

UDP header size and fields?

8 bytes: source port, destination port, length, checksum.

SCENARIO

A live stream drops frames but keeps playing. Protocol?

UDP.

SCENARIO

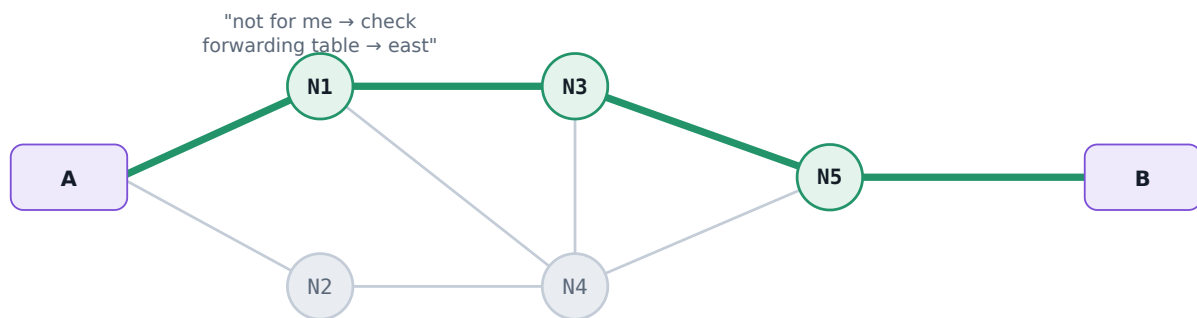
A bank file must arrive complete and ordered. Protocol?

TCP.

Network layer: routing and IP addresses

Routers pass packets hop by hop until they reach the right network. Every packet carries the sender's and receiver's IP address the whole way.

Hop-by-hop forwarding



ROUTING TABLE

Many possible paths

The bigger table

For deciding

FORWARDING TABLE

One chosen path

Part of the routing table

For fast forwarding

CONTROL PLANE · 3:21:50

Builds the routing tables. **Static routing**: routes typed in by hand, not adaptive. **Dynamic routing**: tables update themselves using algorithms like **Bellman-Ford** and **Dijkstra**, the same graph algorithms from DSA.



Your home router doesn't carry strangers' traffic. Hopping happens across ISPs' routers. Their tables hold **blocks** of addresses (subnets) assigned to ISPs like Airtel, Jio and Tata, not every device.

IPv4 address, bit by bit

32 bits written as 4 numbers of 8 bits each (0-255). That gives 2

32

≈

4.3 billion

addresses. Type any address below to see its bits.

IP / PREFIX

192.168.1.0/24

192.168.1.0/24

12.0.0.0/31

127.0.0.1/8

5.6.9.14

192

168

1

0

1 1 0 0 0 0 0 0 0 1 0 1 0 1 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0

☒ network (subnet) bits

☐ host bits

CLASS

C

PREFIX

/24 · 24

network bits, 8

host bits

SUBNET MASK

255.255.255.0

ADDRESSES

256

FIRST

192.168.1.0

LAST

192.168.1.255

TYPE

Private range

Classes, masks and subnets

CLASS	RANGE	DEFAULT MASK (EXTRA)	USE
A	0.0.0.0 – 127.255.255.255	255.0.0.0 · /8	Huge networks
B	128.0.0.0 – 191.255.255.255	255.255.0.0 · /16	Medium
C	192.0.0.0 – 223.255.255.255	255.255.255.0 · /24	Small
D	224.0.0.0 – 239.255.255.255	—	Multicast
E	240.0.0.0 – 255.255.255.255	—	Reserved

SUBNET MASK

Masks the **network** part and leaves the **host** part free. In 192.168.1.1/24 the first 24 bits (192.168.1) are the subnet ID and 1 is the host.

Addresses = $2^{(32 - \text{prefix})}$, so /24 gives 256.

LOOPBACK

127.0.0.0/8 is reserved. 127.0.0.1 is **localhost**: your machine talks to itself, acting as client and server. Always up while the computer runs.



Instructor slips. The video gives Class C's mask as 255.255.0.0; it's **255.255.255.0** (255.255.0.0 is Class B). "First three numbers are network" is only true for Class C. The video credits the IETF with handing out addresses; really **IANA** gives blocks to regional registries (APNIC for India), which give them to ISPs. Early on, big players like MIT, Stanford and IBM got Class A blocks first-come first-served; now allocation is by region.

The IPv4 packet

Header · 20 bytes (extra: up to 60 with options)

Version	Total length	Flags
Identification	TTL	Protocol
Checksum · Source IP · Destination IP		

TTL · TIME TO LIVE

Each hop lowers the TTL. At the limit the packet is **dropped**, so a packet stuck in a loop can't roam forever.

`ping google.com` shows `ttl=60` in the video.

IPv4 vs IPv6 3:41:42

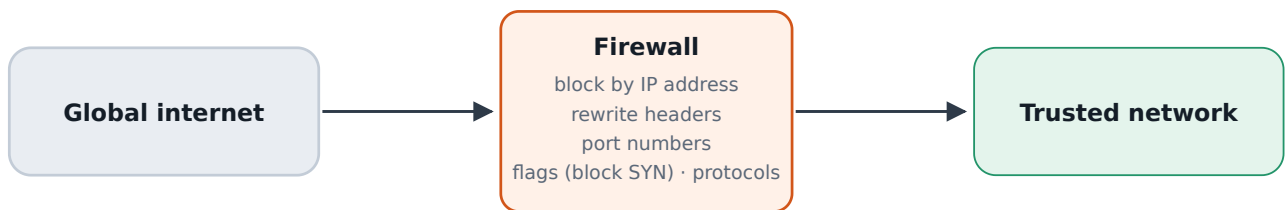
ABFE:F001:3210:9182:0000:0000:0001:0003

shortened → ABFE:F001:3210:9182::1:3

8 groups × 16 bits, in hex. A run of zero groups collapses to :: (extra: only once per address; leading zeros drop).

POINT	IPV4	IPV6
Size	32 bits	128 bits
Looks like	192.168.1.1	ABFE:F001::1:3
Addresses	$2^{32} \approx 4.3$ billion	$2^{128} \approx 3.4 \times 10^{38}$
Why not everywhere yet	—	Not backward compatible · every ISP must switch · hardware must change

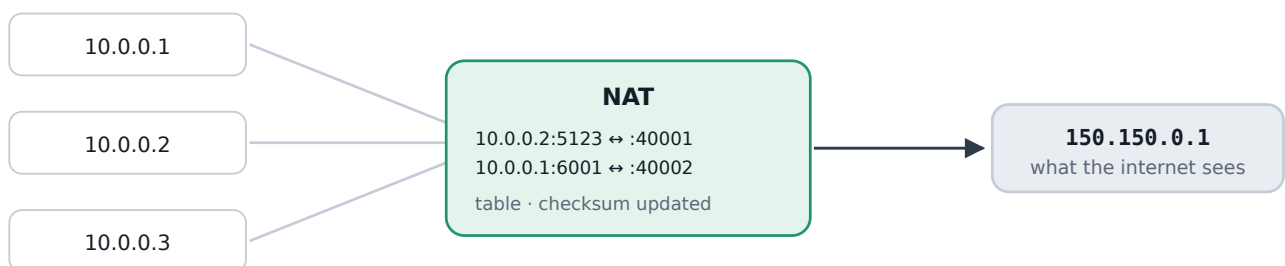
Middle boxes 3:49:50



Middle boxes are devices other than computers and routers that allow, reject or change packets. Mostly at Layer 3, some at Layer 4. Firewalls live on hosts and in the network. Blocking SYN stops anyone opening new connections.

STATELESS FIREWALL	STATEFUL FIREWALL
Judges each packet alone	Remembers packets it has seen (cache)
Less efficient	More efficient

NAT · Network Address Translation 3:52:32



private IP replies come back to 150.150.0.1; NAT looks up its table and forwards to the right private device.

Addresses 10.0.0.x and 150.150.0.1 are the video's example; the port pairs are illustrative. Why NAT exists: to slow the use of the 2^{32} IPv4 addresses (plus security and cost). Also used in enterprise networks and load balancing.

CONCEPT

What is hop-by-hop forwarding?

Each router checks its forwarding table and passes the packet to the next router until it reaches the destination network.

COMPARE

Static vs dynamic routing?

Manual and not adaptive, vs automatic using Bellman-Ford or Dijkstra-style algorithms.

BASIC

How many addresses in a /24?

$2^8 = 256$.

BASIC

What is 127.0.0.1?

Localhost, a loopback address for talking to your own machine.

WHY

Why does TTL exist?

So looping packets are dropped instead of circling forever.

WHY

Why isn't everyone on IPv6?

Not backward compatible; ISPs and hardware must all upgrade.

SCENARIO

How could a firewall block all new incoming connections?

Drop packets carrying the SYN flag.

SCENARIO

Laptop 10.0.0.5 opens google.com. Which IP does Google see?

The NAT's public IP, e.g. 150.150.0.1.

Data link: MAC addresses, ARP and DHCP

IP addresses get a packet to the right network. Inside a LAN, devices actually talk using **MAC addresses**, sending **frames** over a physical link.

DHCP · NEW DEVICE JOINS

- 1 Device connects to the router
- 2 Asks the DHCP server for an IP
- 3 Server picks one from its **pool**
- 4 Device now has its IP

MAC ADDRESS

Media Access Control: a 12-digit alphanumeric ID for one **network interface**. Your Wi-Fi and Bluetooth each have their own. Extra: 48 bits in hex, e.g. `3C:22:FB:9A:1E:07`.

You can block or allow devices by MAC. See yours with `ifconfig`.

ARP: finding a MAC from an IP

① broadcast to everyone

"Who has 192.168.1.4? Tell device 1"

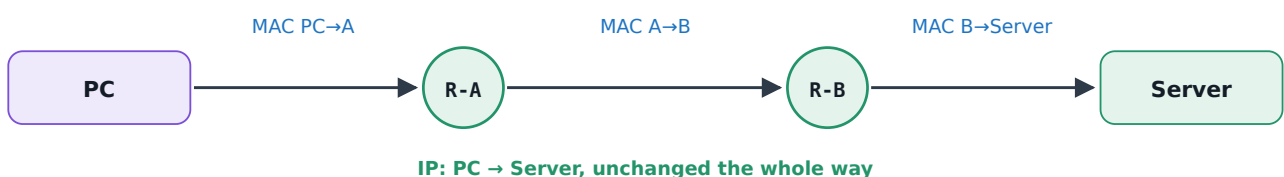
② reply to device 1 only

"192.168.1.4 is at dd:..."



③ device 1 saves IP → MAC in its ARP cache, checked first next time

The IP and MAC shown are examples. Extra context: the **ARP cache** is the saved table; the video uses the name for the whole broadcast step.



DATA LINK ALSO DOES

- **Framing:** building and handling frames
- **Error detection**
- Works closely with the physical layer, handing it 0s and 1s
- On the way back, the router maps the private IP to the right device's MAC

PHYSICAL LAYER

Bits become electrical, light or radio signals on cables and air. The video leaves the electronics for hardware study.

Next in the bootcamp: Docker and Kubernetes networking.

Interview questions · 5

BASIC

What is ARP?

Address Resolution Protocol: finds the MAC for a known IP on the LAN by broadcasting and caching the reply.

COMPARE

MAC vs IP?

MAC is physical, per interface, used within a LAN and rewritten each hop. IP is logical and stays the same end to end.

CONCEPT

How does a new device get an IP?

DHCP assigns one from its pool.

CONCEPT

Does a laptop have one MAC?

No, one per network interface (Wi-Fi, Bluetooth, Ethernet).

SCENARIO

Only known laptops should join your Wi-Fi.

Use MAC address filtering.

R1 Revision

Interview flashcards

Tap a card to flip it. Filter by chapter, or shuffle for a mock interview. Keyboard:

← → to move, **Space** to flip.

Numbers to memorise

OSI / TCP/IP

7 / 5

layers

IPV4

32 bits ≈ 4.3 billion
addresses

IPV6

128 bits $\approx 3.4 \times 10^{38}$
addresses

PORTS

16 bits65,536 ·
reserved 0-1023

UDP HEADER

8 B4 fields × 2
bytes

IPV4 HEADER

20 B

minimum

/24 SUBNET

256addresses (8
host bits)

PING PACKET

64 BTTL ≈ 60 in the
demo

PORT	PROTOCOL / APP	SOURCE
80	HTTP	video
23	Telnet	video
110	POP3	video
27017	MongoDB	video
1433	Microsoft SQL Server	video
443 · 22 · 25 · 53 · 143	HTTPS · SSH · SMTP · DNS · IMAP	extra

Sputnik **1957**1 Mbps = **10^6** bits/scable $\approx$ **28,000 km**Class A-E: **0 · 128 · 192 · 224 · 240**loopback **127.0.0.0/8**handshake **32 → 56/33 → 33/57**NAT **10.0.0.x → 150.150.0.1**MAC **12** hex digits

Pairs people mix up

Internet vs WWW

Network of networks, vs linked pages on top of it.

HTTP vs TCP

HTTP (L7) defines the request and response. TCP (L4) delivers the bytes. HTTP runs on TCP.

Transport vs network layer

Inside a device to the right app, vs between devices across networks.

Segment · packet · frame · bits

● L4 ● L3 ● L2 ● L1

MAC vs IP

Physical, per interface, rewritten each hop, vs logical, end to end. ARP links them.

IP vs port

Which device, vs which app on it.

Public vs private IP

The router's world-facing address, vs DHCP-given inside addresses. NAT translates.

Hub · switch · router

L1 floods everyone · L2 forwards by MAC · L3 forwards by IP between networks.

Checksum · timer · sequence

Corruption · loss · order and duplicates.

Seq vs ack number

Labels my segment, vs the other side's last seq + 1.

POP3 vs IMAP · SMTP

POP3 downloads to one device, IMAP syncs all. SMTP only sends.

Mbps vs MBps

Megabits vs megabytes: divide by 8.

Revision checklist

Tick topics as you master them. Progress shows in the top bar and is remembered in this browser only.

- | | |
|---|---|
| ☐ Network and internet definitions | ☐ History: Sputnik, ARPA, ARPANET |
| ☐ Why protocols exist · RFCs | ☐ WWW, first website, search engines |
| ☐ Client and server · localhost | ☐ Packets |
| ☐ IP address format and phone-book analogy | ☐ Global vs local IP · DHCP · NAT |
| ☐ Ports: 16-bit, ranges, 80/27017/1433 | ☐ Mbps vs MB/s · upload/download |
| ☐ Guided vs unguided media | ☐ Submarine cables |
| ☐ LAN, MAN, WAN · SONET · Frame Relay | ☐ Modem vs router · ISP tiers |
| ☐ 5 topologies and their limits | ☐ Amazon analogy |
| ☐ OSI: all 7 layers | ☐ Encapsulation and the OSI data flow |
| ☐ TCP/IP model · OSI vs TCP/IP | ☐ Data centers · ping |
| ☐ Peer-to-peer · BitTorrent | ☐ Devices: repeater, hub, bridge, switch, router, gateway, brouter |
| ☐ Application protocols and ports | ☐ Program, process, thread |
| ☐ Sockets · ephemeral ports | ☐ HTTP: stateless, over TCP, URL anatomy |
| ☐ HTTP methods and headers | ☐ Status codes 1xx-5xx |
| ☐ Cookies · third-party cookies | ☐ Email: SMTP flow |
| ☐ POP3 vs IMAP | ☐ DNS: domain parts, root, TLD, ICANN |
| ☐ DNS resolution steps · dig | ☐ Transport vs network layer |
| ☐ Multiplexing / demultiplexing | ☐ Congestion control |

- ☐ Checksum · timers · sequence numbers
 - ☐ TCP features
 - ☐ Hop-by-hop forwarding · routing vs forwarding table
 - ☐ IPv4 bits · subnetting · CIDR
 - ☐ IPv4 header · TTL
 - ☐ Firewalls: stateless vs stateful
 - ☐ DHCP pool · ARP
 - ☐ Physical layer signals
 - ☐ UDP header and use cases
 - ☐ 3-way handshake with numbers
 - ☐ Control plane · static vs dynamic routing
 - ☐ IP classes · subnet mask · loopback
 - ☐ IPv6 notation and adoption problems
 - ☐ NAT
 - ☐ MAC addresses · frames
-

What's covered, and what the video got wrong

Built from the video's full auto-generated transcript, its chapter list and Kunal's own notes. Things shown only on screen weren't checked frame by frame.

WHERE	VIDEO SAYS	CORRECT
Ch 1	First ARPANET sites incl. MIT; TCP from the start	UCLA, SRI, UCSB, Utah; NCP first, TCP/IP from 1983
Ch 1	Yahoo was the first search engine ("I believe")	Archie and ALIWEB came earlier; Yahoo was first to go big
Ch 2	NAT = network access translator	Network Address Translation
Ch 4	MAC decides which application	Port picks the app; MAC picks the interface
Ch 6	DHCP "control"; SSH "Secure Socket Shell"	Configuration; Secure Shell
Ch 9	Checksum is random	Calculated from the data
Ch 9	Server seq from client's; 3rd message has SYN	Each side random; 3rd is ACK only
Ch 10	Class C mask 255.255.0.0	255.255.255.0
Ch 10	IETF assigns IPs to ISPs	IANA → regional registries (APNIC) → ISPs

Video coverage checklist · 11 chapters, 40 timestamps

—

Every chapter in the video's timestamp list is covered here, and the promotional segments are skipped. These parts couldn't be verified:

- Exact rows in the DevTools Network demo (shown on screen only)
- Exact command text heard as `curl ifconfig.me -s`
- Which Indian company controls submarine cable access (the instructor was unsure)
- Root-server counts on the map
- The IPv6 prefix example drawn hesitantly on the whiteboard
- The NAT types page scrolled past as "out of scope"

Sources: [Computer Networking Full Course, Kunal Kushwaha \(YouTube\)](#) · [Kunal's networking notes \(GitHub\)](#) · [Networking-Course notes](#)